

FOOD SCIENCE (AGLS)

Food science is a degree program focused on the role of the food industry in producing safe, nutritious, and acceptable foods for consumers. Food scientists apply basic science (chemistry, biology, physics, statistics) to improve processing, preservation, and safety of food and to develop new food products. The food science major is approved by the Institute of Food Technologists.

Student Learning Outcomes

Upon graduation, students should be able to:

- Communicate effectively in their field of study using written, oral, visual and/or electronic forms.
- Demonstrate proficiency in ethical data collection and interpretation, literature review and citation, critical thinking and problem solving.
- Participate effectively in a group or team.
- Integrate creativity, innovation, or entrepreneurship in ways that produce value.
- Explain how human activities impact the natural environment and how societies are affected.
- Meet program specific learning outcomes for the Food Science major.

The department also offers food science minors. (<https://catalog.iastate.edu/collegeofagricultureandlifesciences/foodscienceandhumannutrition/#undergraduateminortext>)

Administered by the Department of Food Science and Human Nutrition

Courses listed below are required.

Total Degree Requirement: 120 cr.

Students must fulfill International Perspectives and U.S. Cultures and Communities requirements by selecting coursework from approved lists. These courses may also be used to fulfill other area requirements. Only 65 cr. from a two-year institution may apply to the degree which may include up to 16 technical cr.; 9 P-NP cr. of electives; 2.00 minimum GPA.

International Perspectives: 3 cr.

U.S. Cultures and Communities: 3 cr.

Communications and Library: 10 cr.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
LIB 1600	Introduction to College Level Research	1
SPCM 2120	Fundamentals of Public Speaking	3
Total Credits		10

Humanities and Social Sciences: 6-12 cr.

Select Humanities course from approved list	3
Social Science course	3

If HSCI student, select: 6

Additional Humanities course

Additional Humanities or Social Science course

Ethics: 3 cr.

FSHN 3420	World Food Issues: Past and Present	3
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Mathematical Sciences: 7-8 cr.

Select 4 credits from:

MATH 1600	Survey of Calculus	4
or MATH 1650	Calculus I	

Select at least 3 credits from:

STAT 1010	Principles of Statistics	3-4
or STAT 1040	Introduction to Statistics	

Total Credits **7-8**

Physical Sciences: 17-19 cr.

CHEM 1770	General Chemistry I	5
& 1770L	and Laboratory in General Chemistry I	
CHEM 1780	General Chemistry II	3
CHEM 2310	Elementary Organic Chemistry	4-6
& 2310L	and Laboratory in Elementary Organic Chemistry	
or CHEM 3310	Organic Chemistry I	
& CHEM 3320	and Organic Chemistry II	
PHYS 1150	Physics for the Life Sciences	5
& 1150L	and Laboratory in Physics for the Life Sciences	
or PHYS 1310	General Physics I	
& 1310L	and General Physics I Laboratory	

Total Credits **17-19**

Biological Sciences: 10-11 cr.

BBMB 3010	Survey of Biochemistry	3
or BBMB 3030	General Biochemistry	
or BBMB 3160	Principles of Biochemistry	
BIOL 2120	Principles of Biology II	3
BIOL 2120L	Principles of Biology Laboratory II	1
MICRO 2010	Introduction to Microbiology	2-3
or MICRO 3020	Biology of Microorganisms	
MICRO 3000L	Introductory Microbiology Laboratory	1

Total Credits **10-11**

Food Science and Human Nutrition: 49 cr.

FSHN 1010	Food and the Consumer	3
FSHN 1100	Professional and Educational Preparation	1
FSHN 1670	Introductory Human Nutrition and Health	3
FSHN 2030	Contemporary Issues in Food Science and Human Nutrition	1

FSHN 2070	Processing of Foods: Basic Principles and Applications	3	ECON 2350	Introduction to Agricultural Markets
FSHN 3050	Food Quality Management and Control	2	ECON 3010	Intermediate Microeconomics
FSHN 3110	Food Chemistry	3	ECON 3200	Labor Economics
FSHN 3110L	Food Chemistry Laboratory	1	ECON 3370	Agricultural Marketing
FSHN 3140	Professional Development for Culinary Food Science and Food Science Majors	1	ECON 3620	Applied Ethics in Agricultural Business
FSHN 3150	Professional Skills for Culinary Food Science and Food Science Majors	1	ECON 4600	Agricultural, Food, and Trade Policy
FSHN 3510	Introduction to Food Engineering Concepts	3	ENTSP 3100	Entrepreneurship and Innovation
FSHN 4030	Food Laws and Regulations	2	FSHN 1150	Food Preparation Laboratory
FSHN 4060	Sensory Evaluation of Food	3	FSHN 1200	The Biochemistry of Beer
FSHN 4070	Microbiological Safety of Foods of Animal Origins	3	FSHN 1200L	Biochemistry of Beer Laboratory
FSHN 4100	Food Analysis	3	FSHN 2140	Scientific Study of Food
FSHN 4110	Food Ingredient Interactions and Formulations	2	FSHN 2150	Advanced Food Preparation Laboratory
FSHN 4120	Food Product Development	3	FSHN 2200	American Food and Culture
FSHN 4200	Food Microbiology	3	FSHN 2420	The US Food System
FSHN 4210	Food Microbiology Laboratory	3	FSHN 2650	Metabolism, Nutrition and Health
FSHN 4710	Food Processing	3	FSHN 2760	Grape and Wine Science
FSHN 4720	Food Processing Laboratory	2	FSHN 4350	Analysis of Food Markets
Total Credits		49	FSHN 4420	Food, Culture, and Sustainability
Select 5-6 credits from the following Professional Electives:		5-6	FSHN 4600	Global Nutrition, Health and Sustainability
ABE 1600	Systematic Problem Solving and Computer Programming		FSHN 4760X	Science and Practice of Grape and Wine
ABE 2160	Fundamentals of Agricultural and Biosystems Engineering		FSHN 4770X	Dairy Products and Dairy Processing
ABE 3250	Biorenewable Systems		FSHN 4900B	Independent Study: Food Science
ABE 4510	Food and Bioprocess Engineering		FSHN 4910B	Supervised Work Experience: Food Science
ACCT 2150	Legal Environment of Business		FSHN 4910D	Supervised Work Experience: Culinary Science
ACCT 2840	Financial Accounting		FSHN 4990	Undergraduate Research
ACCT 2850	Managerial Accounting		FSHN 4960A	Food Science and Human Nutrition Travel Course: International travel
AESHM 4740	Entrepreneurship in Human Sciences		GLOBE 2010	Introduction to Global Resource Systems
AGRON 4500	Issues in Sustainable Agriculture		GLOBE 2200	Globalization and Sustainability
ANS 2700	Foods of Animal Origin		GLOBE 3030	Agricultural, Food and Natural Global Resource Systems
ANS 2700L	Foods of Animal Origin Laboratory		HORT 2210	Principles of Horticulture Science
ANS 3600	Fresh Meat Science and Applied Muscle Biology		HORT 4610	Fruit Crop Production and Management
ANS 4600	Science and Technology of Value Added Meat Products		HORT 4710	Vegetable Production and Management
CHEM 2110	Quantitative and Environmental Analysis		HORT 4710L	Vegetable Production and Management Lab
CHEM 2110L	Quantitative and Environmental Analysis Laboratory		HSPM 1330	Food Safety Certification
CHEM 3160	Instrumental Methods of Chemical Analysis		MGMT 3710	Organizational Behavior
CHEM 3160L	Instrumental Analysis Laboratory		MGMT 4140	International Management
			MGMT 4720	Building Inclusive Work Cultures
			MIS 3010	Management Information Systems
			MKT 3400	Principles of Marketing
			MKT 4470	Consumer Behavior

MKT 4480	Global Marketing	
Total Credits		5-6

Electives: 0-10 cr. Select from any university coursework to earn at least 120 total credits. Food science internship experience is strongly recommended during the summers, and students can earn elective credits for the internship experience by enrolling in FSHN 4910B.

Go to FSHN courses. (https://catalog.iastate.edu/azcourses/fs_hn/)

Food Science, B.S.

First Year

Fall	Credits Spring	Credits
FSHN 1010	3 FSHN 1670	3
FSHN 1100	1 CHEM 1780	3
CHEM 1770	4 BIOL 2120	3
CHEM 1770L	1 BIOL 2120L	1
ENGL 1500	3 Humanities	3
LIB 1600	1 Social Science	3
Humanities or Elective	3	
	16	16

Second Year

Fall	Credits Spring	Credits
CHEM 2310 and CHEM 2310L or CHEM 3310	3-4 BBMB 3010, 3030, or 3160	3
FSHN 2030	1 CHEM 3320 (if CHEM 3310 taken) or Elective	3
ENGL 2500	3 MICRO 2010 or 3020	2-3
PHYS 1150 or 1310	4 MICRO 3000L	1
PHYS 1150L or 1310L	1 STAT 1010 or 1040	3-4
MATH 1600 or 1650	4 FSHN 2070	3
	16-17	15-17

Third Year

Fall	Credits Spring	Credits
FSHN 3110	3 FSHN 3050	2
FSHN 3110L	1 FSHN 3510	3
FSHN 3140	1 FSHN 4030	2
FSHN 3150	1 FSHN 4110	2
FSHN 4200	3 FSHN 4210	3
SPCM 2120	3 Professional Elective	3
Professional Elective	3	
	15	15

Fourth Year

Fall	Credits Spring	Credits
FSHN 4060	3 FSHN 3420	3
FSHN 4100	3 FSHN 4120	3

FSHN 4710	3 FSHN 4720	2
FSHN 4070	3 U.S. Cultures and Communities (if not already taken) or elective	3
Elective *	2 Humanities/social science (CHHS) or elective (CALS)	3
	14	14

* Choose elective courses to total equal to or greater than 120 credits.

This sequence is only an example. The number of credits taken each semester should be based on the individual student's situation. Factors that may affect credit hours per semester include student ability, employment, health, activities, and grade point considerations.

More information on the Food Science, Food Safety, and Food and Society minors can be found here: <http://catalog.iastate.edu/collegeofagricultureandlifesciences/foodscienceandhumannutrition/#undergraduateminortext> (<https://catalog.iastate.edu/collegeofagricultureandlifesciences/foodscienceandhumannutrition/#undergraduateminortext>).

The Department of Food Science and Human Nutrition offers a Master of Science (M.S.) and Doctor of Philosophy (Ph.D.) in Food Science and Technology. More information can be found here: <https://www.grad-college.iastate.edu/academics/programs/apresults.php?id=50>.